

[DHS/2025/Q1]

- 1 Dunman High School wants to store records of its IT Devices in a NoSQL database. The list of devices (laptops & tablets) are recorded in the file `IT_Devices.json`.

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [ ]: #Task 1.1
        Program code
```

Output:

Task 1.1

Write program code to insert the data from the file `IT_Devices.json` into a NoSQL database `DHS` under the collection `IT_Devices`. The program should clear the `IT_Devices` collection if it exists.

(Note: You may refer to the `read_json_sample.py` file for an example of how to extract data from JSON files.)

Retrieve and display all the documents in the collection `IT_Devices` after the insertion of data. [4]

Task 1.2

The school has decided to change all occurrences of the brand name `HP` to its full name, `Hewlett-Packard`. Write a program to replace all values of `HP` in the collection `IT_Devices` with `Hewlett-Packard`. [3]

Subsequently, extend the above program to retrieve and display all laptops that weigh less than 2 kg together with all tablets that have at least 8000 mAh Battery Capacity. Display all relevant devices in the following format: [4]

```
Serial No: LAP-001, Type: Laptop, Brand: Hewlett-Packard, Weight: 1.35
Serial No: TAB-002, Type: Tablet, Brand: Apple, Battery Capacity: 9720
```

Task 1.3

The school decided to move all of its data into a relational database (`DHS.db`). The database has three tables: one table to store the devices' common information, and separate tables for laptops and tablets to store their specific information.

```
Device (Serial_Number, Type, Brand, Cost, Date_Of_Purchase)
```

```
Laptop (Serial_Number, Weight, Screen_Size)
```

```
Tablet (Serial_Number, Battery_Capacity)
```

Device:

- `Serial_Number` – the unique device identifier, for example, `LAP-001`
- `Type` – the category of the device, either `Laptop` or `Tablet`
- `Brand` – the brand of the device, for example, `Dell`
- `Cost` – the cost of the device, for example, `2199`
- `Date_Of_Purchase` – the purchase date of the device in `YYYY-MM-DD`

Laptop:

- `Serial_Number` – the unique device identifier, for example, `LAP-001`
- `Weight` – the weight of the laptop in kg, for example, `2.6`
- `Screen_Size` – the screen size of the laptop in inches, for example, `17`

[DHS/2025/Q1 Cont']

Tablet:

- `Serial_Number` – the unique device identifier, for example, `TAB-001`
- `Battery_Capacity` – the battery capacity of the tablet in mAh, for example, 7500

Write a Python program that uses SQL code to create the database `DHS.db` with the three tables given. Define the primary and foreign keys for each table. [6]

Additionally, extend the above program to retrieve all data from the DHS NoSQL database's collection, `IT_Devices`, and store the data in the correct place in the relational database, `DHS.db`. [5]

Task 1.4

The school would like to keep a log of all the school devices that are older than 1000 days old.

Write program code to

- extract relevant data from the relational database, `DHS.db`
- calculate the age of each device (in days) by subtracting the purchase date from today's date
- write the `Serial_Number`, `Type`, `Date_Of_Purchase` and age of device (in days) of devices older than 1000 days old to a csv file with the name `Old_devices.csv`

The `datetime` library is built into Python and can be used to manipulate dates. See the example code shown in `datetime_sample.py` and use the `datetime` library to calculate the age of device (in days). [6]

Run your program.

Save your Jupyter Notebook for Task 1.